University of New Haven

TAGLIATELA COLLEGE OF ENGINEERING

POWER ON

Presented by :
Team 1

Scalable Ride Demand Prediction using a Distributed ETL Pipeline.

Ridepulse.

Pitch Deck
Presentation
DCSI 6007

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Today's Agenda

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10 Expected outcome & Business use cases

11 Final Project preview

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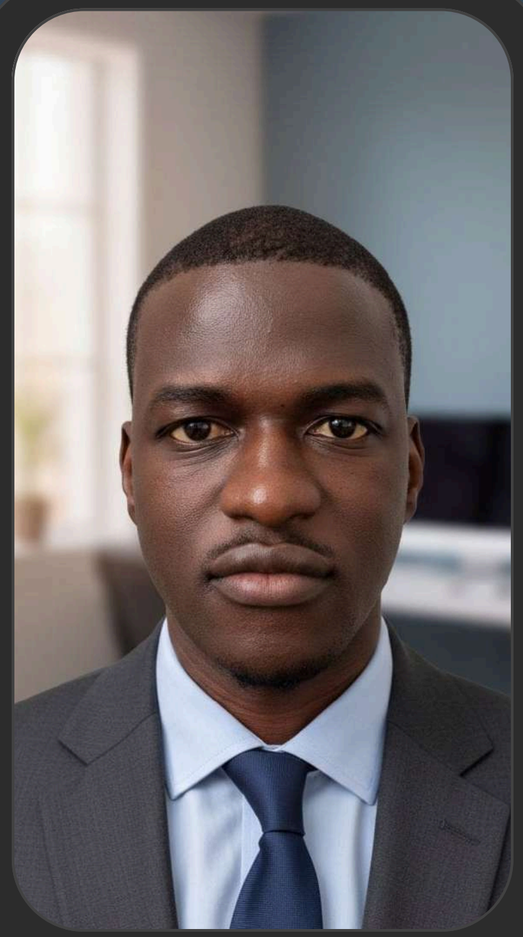
Meet Our Team

HARSHAVRDHAN J

TEAM LEADER

DATA ENGINNER

DATA SCIENTIST



CUTHBETH MASVAURE

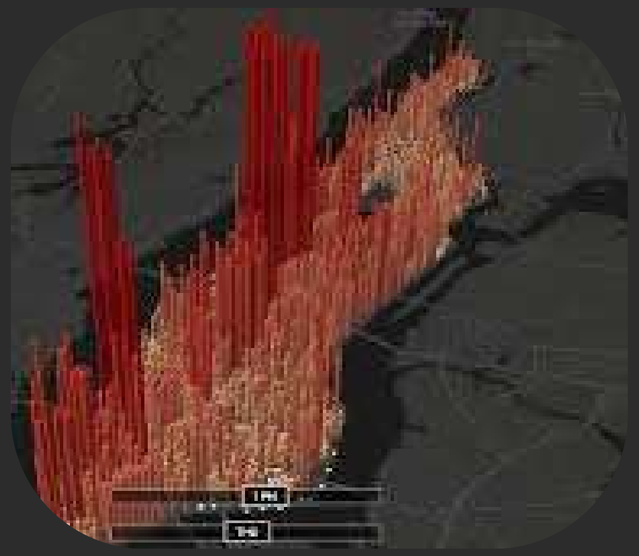
DATABASE ARCHITECT

VISUALIZATION & STORYTELLING LEAD

Introduction

Urban mobility platforms generate massive amounts of ride data every day. Efficiently processing and analyzing this data is essential for optimizing driver allocation and reducing passenger wait times.

This project focuses on building a scalable ETL pipeline to process large-scale ride data and support demand prediction using distributed data engineering principles.



Problem Statement

- DEMAND SPIKES AT PEAK HOURS
- POOR RESOURCE ALLOCATION
- SURGE PRICING INEFFICIENCIES
- DELAYS IN HIGH-DEMAND AREAS

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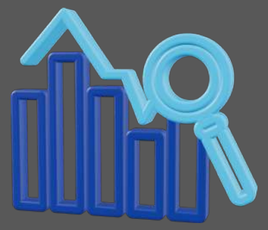
Companies need:

Real-time data ingestion

Scalable processing



Demand prediction

Actionable dashboards

Proposed Solution: (Midterm phase 1)

We will build:
A Distributed & Scalable ETL Pipeline that:

1. Ingests ride data(NYC Taxi Dataset or similar)

2. Cleans and transforms data

3. Stores it in a data warehouse

4. Builds demand prediction model

5. Visualizes demand trends

ETL Architecture Diagram

Conceptual Architecture

CRISP-DM Framework

1. Business Understanding

Predict hourly NYC taxi demand
Improve driver allocation & efficiency

2. Data Understanding

- NYC Yellow Taxi dataset
- Trip-level time-series data
- Identified peak-hour patterns

3. Data Preperation

- Cleaned invalid trips
- Removed outliers
- Engineered time-based features
- Aggregated hourly demand

4. Modeling

Linear Regression (baseline)
Random Forest (improved model)
Time & peak-hour features

5. Evaluation

- RMSE ≈ 1746 (~10% error)
- Model captures rush-hour spikes

6. Deployment (Planned)

- ETL pipeline built
- Stored in database
- Future: Spark + Cloud deployment

Expected Outcome

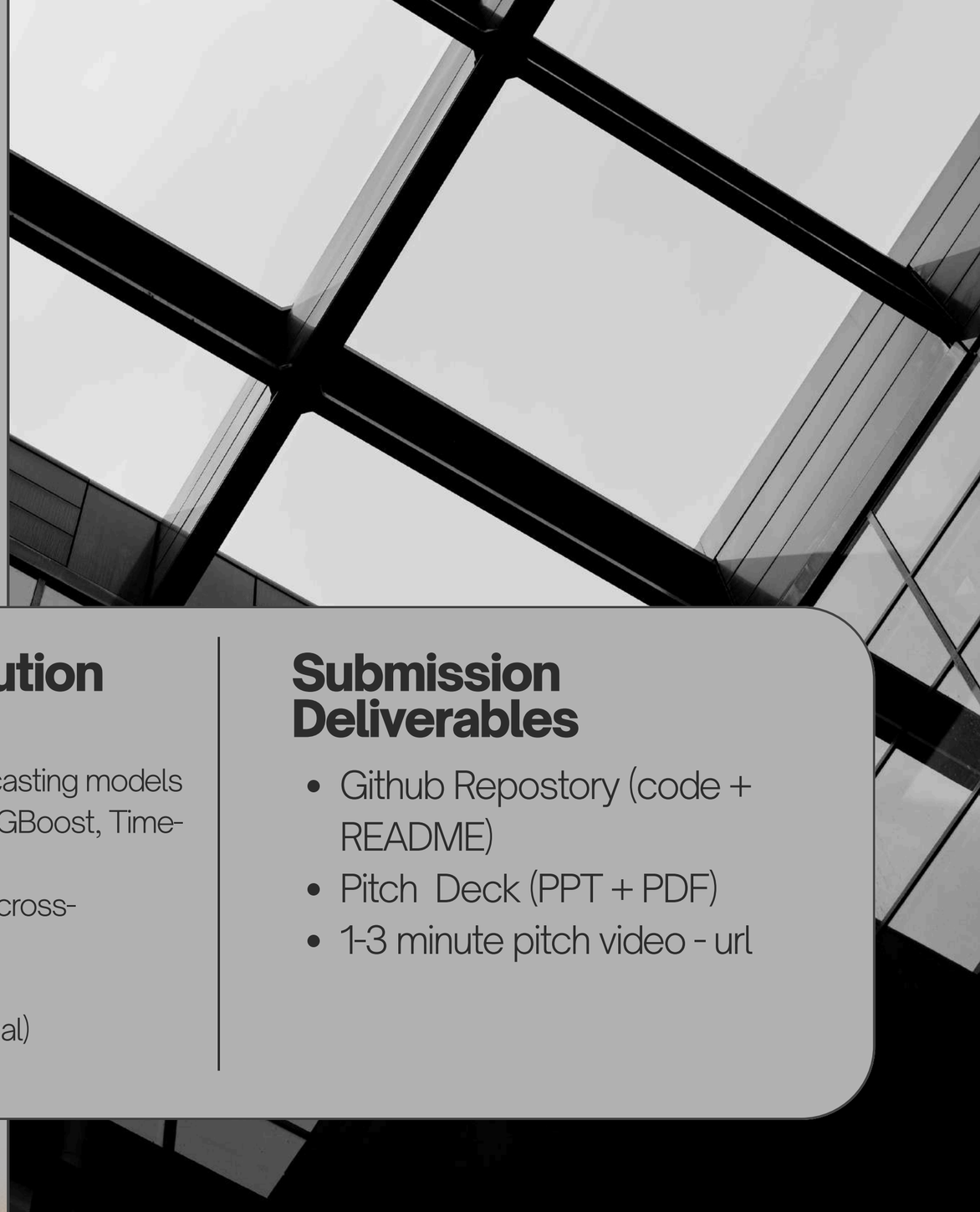
- Defined project Scope + Dataset schema
- designed Scalable ETL architecture diagram
- Implemented CRISP-DM framework
- Built local ETL + modeling prototype
- Stored aggeregated data in database

Business Use Cases

- Ride-sharing platforms: optimize driver allocation
- Reduce passenger wait time
- Identify peak-hour demand patterns
- Support surge pricing strategy
- Future: real-time demand forecasting

Final Project Preview

After midterm approval we will scale the pipeline with stronger engineering & model deployment



Engineernig Upgrades

* Replace batch loading with Spark processing
* Implement distributed ETL pipeline
* Move storage to PostgreSQL / Cloud warehouse
* Add data validation & monitoring
* Optional: Kafka streaming ingestion

Modeling & Evalution Upgrades

Train advanced demand forecasting models
* Compare Random Forest, XGBoost, Time-Series
* Evaluate using RMSE + R^2 + cross-validation
* Monitor model drift
* Deploy model as API (optional)

Submission Deliverables

- Github Repostory (code + README)
- Pitch Deck (PPT + PDF)
- 1-3 minute pitch video - url

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Thank You

for your time
and attention